

## Joshua J. Goings, Ph.D.

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|------------------|-------|-----------------------------------------|--------------------------|------|
| <b>EDUCATION</b> | Ph.D. | University of Washington, Seattle, WA   | Theoretical Chemistry    | 2017 |
|                  | B.S.  | Seattle Pacific University, Seattle, WA | Chemistry & Biochemistry | 2012 |

### EXPERIENCE

|                                                                                                                                                                |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Staff Applications Engineer<br>IonQ                                                                                                                            | 02/2022 to present |
| Visiting Research Scientist<br>Google Quantum AI                                                                                                               | 07/2021 to 02/2022 |
| Data Science Postdoctoral Research Fellow<br>eScience Institute, University of Washington                                                                      | 03/2021 to 07/2021 |
| Postdoctoral Research Associate<br>University of Washington<br>Advisor: Prof. Xiaosong Li                                                                      | 07/2020 to 07/2021 |
| Postdoctoral Research Associate<br>Yale University (lab moved Jan 2018)<br>University of Illinois at Urbana-Champaign<br>Advisor: Prof. Sharon Hammes-Schiffer | 08/2017 to 06/2020 |
| Graduate Research Assistant<br>NSF Graduate Research Fellow<br>University of Washington<br>Advisor: Prof. Xiaosong Li                                          | 07/2012 to 06/2017 |

### PUBLICATIONS & PREPRINTS

- 43 **Measuring what matters: A scalable framework for application-level quantum benchmarking**  
W. Aboumrاد, C. Girotto, J. Goings, L. Zhao, M. A. Lopez-Ruiz, D. Zhu, A. Kaushik, S. Ray, S. Sekwao, J. Iaconis, et al.  
*arXiv preprint arXiv:2604.11781*, 2026.
- 42 **Molecular properties in quantum-classical auxiliary-field quantum Monte Carlo: Correlated sampling with application to accurate nuclear forces**  
J. J. Goings, K. Shin, S. Noh, W. Kyoung, D. Kim, J. Baek, M. Roetteler, E. Epifanovsky, L. Zhao  
*J. Chem. Theory Comput.*, 2025, 21, 12573–12591.
- 41 **Quantum-classical auxiliary field quantum Monte Carlo with matchgate shadows on trapped ion quantum computers**  
L. Zhao, J. J. Goings, W. Aboumrاد, A. Arrasmith, L. Calderin, S. Churchill, D. Gabay, T. Harvey-Brown, M. Hiles, M. Kaja, et al.  
*arXiv preprint arXiv:2506.22408*, 2025.
- 40 **Enhancing the electron pair approximation with measurements on trapped-ion quantum computers**  
L. Zhao, Q. Wang, J. J. Goings, K. Shin, W. Kyoung, S. Noh, Y. M. Rhee, K. Kim  
*npj Quantum Inf.*, 2024, 10, 76.
- 39 **Quantum algorithm exploration using application-oriented performance benchmarks**  
T. Lubinski, J. J. Goings, K. Mayer, S. Johri, N. Reddy, A. Mehta, N. Bhatia, S. Rappaport, D.

Mills, C. H. Baldwin, et al.  
*arXiv preprint* arXiv:2402.08985, 2024.

- 38 **Molecular symmetry in VQE: A dual approach for trapped-ion simulations of benzene**  
J. Goings, L. Zhao, J. Jakowski, T. Morris, R. Pooser  
*IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2023, 76–82.
- 37 **Orbital-optimized pair-correlated electron simulations on trapped-ion quantum computers**  
L. Zhao, J. Goings, K. Shin, W. Kyoung, J. I. Fuks, J.-K. Kevin Rhee, Y. M. Rhee, K. Wright, J. Nguyen, J. Kim, S. Johri  
*npj Quantum Inf.*, 2023, 9, 60.
- 36 **Pushing the limits of quantum computing for simulating PFAS chemistry**  
E. Dimitrov, G. Sanchez-Sanz, J. Nelson, L. O’Riordan, M. Doyle, S. Courtney, V. Kannan, H. Naseri, A. G. Garcia, J. Tricker, ...J. Goings, et al.  
*arXiv preprint* arXiv:2311.01242, 2023.
- 35 **Reliably assessing the electronic structure of cytochrome P450 on today’s classical computers and tomorrow’s quantum computers**  
J. J. Goings, A. White, J. Lee, C. S. Tautermann, M. Degroote, C. Gidney, T. Shiozaki, R. Babbush, N. C. Rubin  
*Proc. Natl. Acad. Sci. U.S.A.*, 2022, 119, e2203533119.
- 34 **Exploring potential energy surfaces using reinforcement machine learning**  
A. W. Mills, J. J. Goings, D. Beck, C. Yang, X. Li  
*J. Chem. Inf. Model.*, 2022, 62, 3169–3179.
- 33 **Efficient quantum computation of molecular forces and other energy gradients**  
T. E. O’Brien, M. Streif, N. C. Rubin, R. Santagati, Y. Su, W. J. Huggins, J. J. Goings, N. Moll, E. Kyoseva, M. Degroote, C. S. Tautermann, J. Lee, D. W. Berry, N. Wiebe, R. Babbush  
*Phys. Rev. Research*, 2022, 4, 043210.
- 32 **Determination of the SmO<sup>+</sup> bond energy by threshold photodissociation of the cryogenically cooled ion<sup>\*</sup>**  
A. Lachowicz, E. H. Perez, N. S. Shuman, S. Ard, A. A. Viggiano, P. B. Armentrout, J. J. Goings, P. Sharma, X. Li, and M. A. Johnson  
*J. Chem. Phys.*, 2021, 155, 174303.  
<sup>\*</sup>2021 JCP Editors’ Pick
- 31 **Reinforcement Learning Configuration Interaction**  
J. J. Goings, H. Hu, C. Yang, and X. Li  
*J. Chem. Theory Comput.*, 2021, 17, 5482–5491.
- 30 **Role of Intact Hydrogen-Bond Networks in Multiproton-Coupled Electron Transfer**  
W. D. Guerra, E. Odella, M. Secor, J. J. Goings, M. N. Urrutia, B. L. Wadsworth, M. A. Gervaldo, L. E. Sereno, T. A. Moore, G. F. Moore, S. Hammes-Schiffer, A. L. Moore  
*J. Amer. Chem. Soc.* 2020, 142, 21842–21851.
- 29 **Formation of an Unusual Glutamine Tautomer in a Blue Light Using Flavin Photocycle Characterizes the Light-adapted State**  
J. J. Goings, P. Li, Q. Zhu, S. Hammes-Schiffer  
*Proc. Natl. Acad. Sci.* 2020, 117, 26626–26632.
- 28 **Nonequilibrium Dynamics of Proton-Coupled Electron Transfer in Proton Wires: Concerted but Asynchronous Mechanisms**  
J. J. Goings, S. Hammes-Schiffer  
*ACS Cent. Sci.* 2020, 6, 1594–1601.

- 27 **The Chronus Quantum Software Package**  
D. B. Williams-Young, A. Petrone, S. Sun, T. F. Stetina, P. Lestrangle, C. E. Hoyer, D. R. Nascimento, L. Koulias, A. Wildman, J. Kasper, J. J. Goings, F. Ding, A. E. DePrince III, E. F. Valeev, X. Li  
*WIREs Comput. Mol. Sci.* 2020, e1436.
- 26 **Proton-Coupled Electron Transfer Across Benzimidazole Bridges in Bioinspired Proton Wires**  
E. Odella<sup>†</sup>, S. J. Mora<sup>†</sup>, B. L. Wadsworth<sup>†</sup>, J. J. Goings<sup>†</sup>, M. A. Gervaldo, L. E. Sereno, T. L. Groy, D. Gust, T. A. Moore, G. F. Moore, S. Hammes-Schiffer, A. L. Moore  
*Chem. Sci.* 2020, 11, 3820–3828.  
<sup>†</sup>Equal contribution.
- 25 **Early Photocycle of Slr1694 Blue-Light Using Flavin Photoreceptor Unraveled through Adiabatic Excited-State Quantum Mechanical/Molecular Mechanical Dynamics**  
J. J. Goings, S. Hammes-Schiffer  
*J. Amer. Chem. Soc.* 2019, 141, 20470–20479.
- 24 **Proton-Coupled Electron Transfer Drives Long-Range Proton Translocation in Bioinspired Systems**  
E. Odella<sup>†</sup>, B. L. Wadsworth<sup>†</sup>, S. J. Mora<sup>†</sup>, J. J. Goings<sup>†</sup>, M. T. Huynh, D. Gust, T. A. Moore, G. F. Moore, S. Hammes-Schiffer, A. L. Moore  
*J. Amer. Chem. Soc.* 2019, 141, 14057–14061.  
<sup>†</sup>Equal contribution.
- 23 **Electron-Coupled Double Proton Transfer in the Slr1694 BLUF Photoreceptor: A Multireference Electronic Structure Study**  
E. R. Sayfutyarova, J. J. Goings, S. Hammes-Schiffer  
*J. Phys. Chem. B.* 2019, 123, 439–447.
- 22 **Propensity for Proton Relay and Electrostatic Impact of Protein Reorganization in Slr1694 BLUF Photoreceptor**  
J. J. Goings, C. R. Reinhardt, S. Hammes-Schiffer  
*J. Amer. Chem. Soc.* 2018, 140, 15241–15251.
- 21 **Controlling Proton-Coupled Electron Transfer in Bio-Inspired Artificial Photosynthetic Relays**  
E. Odella, S. J. Mora, B. L. Wadsworth, M. T. Huynh, J. J. Goings, P. A. Liddell, T. L. Groy, M. Gervaldo, L. E. Sereno, D. Gust, T. A. Moore, G. F. Moore, S. Hammes-Schiffer, A. L. Moore  
*J. Amer. Chem. Soc.* 2018, 140, 15450–15460.
- 20 **Orientation-Dependent Imaging of Electronically Excited Quantum Dots\***  
D. Nguyen, J. J. Goings, H. A. Nguyen, J. Lyding, X. Li, M. Gruebele  
*J. Chem. Phys.*, 2018, 148, 064701.  
\*2018 JCP Editors' Choice
- 19 **Current Development of Noncollinear Electronic Structure Theory**  
J. J. Goings, F. Egidi, X. Li  
*Int. J. Quantum Chem.* 2018, 118, e25398.
- 18 **Real-Time Time-Dependent Electronic Structure Theory**  
J. J. Goings, P. J. Lestrangle, X. Li  
*WIREs Comput. Mol. Sci.* 2017, e1341.
- 17 **Two-Component Non-Collinear Time-Dependent Spin Density Functional Theory for Excited State Calculations**  
F. Egidi, S. Sun, J. J. Goings, G. Scalmani, M. J. Frisch, X. Li  
*J. Chem. Theory Comput.* 2017, 13, 2591–2603.

- 16 **Can Quantized Vibrational Effects Be Obtained from Ehrenfest Mixed Quantum-Classical Dynamics?**  
J. J. Goings, D. B. Lingerfelt, X. Li  
*J. Phys. Chem. Lett.* 2016, 7, 5193–5197.
- 15 **Accelerating Real-Time Time-Dependent Density Functional Theory with a Non-Recursive Chebyshev Expansion of the Quantum Propagator**  
D. Williams-Young, J. J. Goings, X. Li  
*J. Chem. Theory Comput.* 2016, 12, 5333–5338.
- 14 **Quantum Confinement Effects on Optical Transitions in Nanodiamonds Containing Nitrogen Vacancies**  
A. Petrone, J. J. Goings, X. Li  
*Phys. Rev. B.* 2016, 94, 165402.
- 13 **Real Time Propagation of the Exact Two Component Time-Dependent Density Functional Theory**  
J. J. Goings, J. M. Kasper, F. Egidi, S. Sun, X. Li  
*J. Chem. Phys.* 2016, 145, 104107.
- 12 **A Direct Atomic-Orbital Based Relativistic Two-Component Linear Response Method for Calculating Excited State Fine Structures**  
F. Egidi, J. J. Goings, M. J. Frisch, X. Li  
*J. Chem. Theory Comput.* 2016, 12, 3711–3718.
- 11 **An Atomic Orbital Based Real-Time Time-Dependent Density Functional Theory for Computing Electronic Circular Dichroism Band Spectra**  
J. J. Goings, X. Li  
*J. Chem. Phys.* 2016, 144, 234102.
- 10 **Imaging Excited Orbitals of Quantum Dots: Experiment and Electronic Structure Theory**  
L. Nienhaus<sup>†</sup>, J. J. Goings<sup>†</sup>, D. Nguyen, S. Wieghold, J. Lyding, X. Li, M. Gruebele  
*J. Amer. Chem. Soc.* 2015, 137, 14743–14750.  
<sup>†</sup>Equal contribution.
- 9 **Approximate Singly Excited States from a Two-Component Hartree-Fock Reference**  
J. J. Goings, F. Ding, E. R. Davidson, X. Li  
*J. Chem. Phys.* 2015, 143, 144106.
- 8 **Ab Initio Two-Component Ehrenfest Dynamics**  
F. Ding, J. J. Goings, H. Liu, D. Lingerfelt, X. Li  
*J. Chem. Phys.* 2015, 143, 114105.
- 7 **Energy-Specific Equation-of-Motion Coupled-Cluster Methods for High-Energy Excited States: Application to K-Edge X-ray Absorption Spectroscopy**  
B. Peng, P. J. LeStrange, J. J. Goings, M. Caricato, X. Li  
*J. Chem. Theory Comput.* 2015, 11, 4146–4153.
- 6 **Stability of the Complex Generalized Hartree-Fock Equations**  
J. J. Goings, F. Ding, M. J. Frisch, X. Li  
*J. Chem. Phys.* 2015, 142, 154109.
- 5 **Ab Initio Non-Relativistic Spin Dynamics**  
F. Ding, J. J. Goings, M. J. Frisch, X. Li  
*J. Chem. Phys.* 2014, 141, 214111.

- 4 **Theoretical Characterization of Conduction-Band Electrons in Photodoped and Aluminum-Doped Zinc Oxide (AZO) Quantum Dots**  
J. J. Goings, A. M. Schimpf, J. W. May, R. W. Johns, D. R. Gamelin, X. Li  
*J. Phys. Chem. C* 2014, *118*, 26584–26590.
- 3 **Assessment of Low-Scaling Approximations to the Equation of Motion Coupled-Cluster Singles and Doubles Equations**  
J. J. Goings, M. Caricato, M. Frisch, X. Li  
*J. Chem. Phys.* 2014, *141*, 164116.
- 2 **Sorption of H<sub>2</sub> to Open Metal Sites in a Metal Organic Framework: A Symmetry Adapted Perturbation Analysis**  
J. J. Goings, S. Ohlsen, K. Blaisdell, D. Schofield  
*J. Phys. Chem. A* 2014, *118*, 7411–7417.
- 1 **Self-Consistent-Field Using Direct Inversion in Iterative Subspace Method and Quasi-Newton Vectors**  
J. J. Goings, F. Ding, X. Li  
*Adv. Quantum Chem.* 2014, *68*, 77–86.

#### AWARDS

|                                                            |      |
|------------------------------------------------------------|------|
| University of Washington Data Science Postdoctoral Fellow  | 2021 |
| Chemical Computing Group (CCG) Excellence Award            | 2016 |
| National Science Foundation (NSF) Graduate Research Fellow | 2014 |
| Bernard and Claudine Nist Endowed Fellowship in Chemistry  | 2014 |
| Peter Salamon Award, Telluride Science Research Center     | 2014 |
| Natt-Lingafelter Graduate Fellowship Award                 | 2012 |
| David T. & Christina L. Wong Scholarship in Chemistry      | 2011 |

#### ORAL PRESENTATIONS

- *Molecular symmetry in VQE: A dual approach for trapped-ion simulations of benzene.*  
IEEE International Conference on Quantum Computing and Engineering (QCE), Bellevue, WA (2023)
- *Impact of protein conformational changes and electrostatics on a BLUF photoreceptor.*  
258<sup>th</sup> ACS National Meeting, San Diego, CA (2019)
- *Engineering a redox-active proton wire with proton-coupled electron transfer.*  
258<sup>th</sup> ACS National Meeting, San Diego, CA (2019)
- *Can quantized vibrational effects be obtained from Ehrenfest mixed quantum-classical dynamics?*  
253<sup>rd</sup> ACS National Meeting, San Francisco, CA (2017)
- *Atomic-orbital based real-time TDDFT for circular dichroism spectroscopy.*  
251<sup>st</sup> ACS National Meeting, San Diego, CA (2016)
- *Theoretical investigation of magnetic exchange interactions in dilute magnetic semiconductor quantum dots induced by defects.*  
Pacifichem 2015, Honolulu, HI (2015)
- *Linear Response Complex Generalized Hartree-Fock.*  
249<sup>th</sup> ACS National Meeting, Denver, CO (2015)
- *Equation of motion formalism of second order many-body perturbation theory (EOM-MBPT2) and second-order approximate coupled-cluster (CC2).*  
248<sup>th</sup> ACS National Meeting, San Francisco, CA (2014)

- *Low-Scaling Approximations to the Equation of Motion Coupled-Cluster Singles and Doubles Equations.*  
Excited States and Time-Dependent Electronic Structure Theory, Telluride, CO (2014)

#### POSTER PRESENTATIONS

- *Propensity for Proton Relay and Impact of Protein Reorganization in Slr1694 BLUF Photoreceptor.*  
Gordon Research Conference on Computational Chemistry, West Dover, VT (2018)
- *Real Time Propagation of the Exact Relativistic Two-Component Equations.*  
Theory and Applications of Computational Chemistry (TACC), Seattle, WA (2016)
- *Recent advances in real-time TDDFT for the description of optical activity.*  
Low-scaling and Unconventional Electronic Structure Techniques Conference (LUEST), Telluride, CO (2016)
- *Towards a Real-Time Description of Magnetic Systems with Applications to Magnetic Circular Dichroism Spectroscopy.*  
251<sup>st</sup> ACS National Meeting, San Diego, CA (2016)

#### TEACHING

Lecturer, Chemistry, University of Washington

Winter 2021

*CHEM 465/565: Computational Chemistry*

- Lead instructor; re-designed course for asynchronous online learning.
- Managed team of TAs to facilitate hands-on student experience with modern computational chemistry software and UW's high-performance computing (HPC) resources.
- Lectures highlighted fundamental and recent developments in computational chemistry.
- Guided students in novel research projects utilizing state-of-the art methods and computing resources directly applicable to individual research interests.

Teaching Assistant, Chemistry, Yale University

Spring 2019

*CHEM 426/526: Computational Chemistry and Biochemistry*

- Designed a computational chemistry course for graduate and undergraduate students.
- Gave students hands-on guidance using computational chemistry software.
- Facilitated student access to Yale's high-performance computing center.
- Performed computational "demos" at the end of each lecture; guest lectured as necessary.